IMPACT OF DIFFERENT TYPES OF FERTILIZERS ON CROP YIELD

Minor project-1 report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Computer Science & Engineering

By

G.SOWMYA	(22UECS0243)	**(21917)**
M.BHARGAVI	(22UECS0427)	**(23524)**
V.LOHITHA	(22UECS0746)	**(23504)**

Under the guidance of
Dr.S.Karthiyayini,M.E,Ph.D.,
Assistant Professor Senior Grade.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE & TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
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CERTIFICATE

It is certified that the work contained in the project report titled" THE IMPACT OF DIFFERENT TYPES OF FERTILIZERS ON CROP YIELD" by "G.SOWMYA (22UECS0243) , M.BHARGAVI (22UECS0427) , V.LOHITHA (22UECS0746)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Signature of Supervisor
Dr.S.karthiyayini
Assistant Professor Senior Grade
Computer Science & Engineering
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science & Technology
December, 2024

Signature of Head of the Department
Dr. N. Vijayaraj
Professor & Head
Computer Science & Engineering
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science & Technology
December, 2024

Signature of the Dean
Dr. S P. Chokkalingam
Professor & Dean
Computer Science & Engineering
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science & Technology
December, 2024

DECLARATION

We declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Signature)
G.SOWMYA
Date: / /

(Signature)
V.LOHITHA
Date: / /

(Signature)
M.BHARGAVI
Date: / /

APPROVAL SHEET

This project report entitled"IMPACT OF DIFFERENT TYPES OF FERTILIZERS ON CROP YIELD" by G.SOWMYA(22UECS0243) , M.BHARGAVI(22UECS0427) , V.LOHITHA(22UECS0746) is approved for the degree of B.Tech in Computer Science & Engineering.

Examiners **Supervisor**

Dr.S.Karthiyayini,M.E,Ph.D.,

Date: / /

Place:

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We express our deepest gratitude to our **Honorable Founder Chancellor and President Col. Prof. Dr. R. RANGARAJAN B.E. (Electrical), B.E. (Mechanical), M.S (Automobile), D.Sc., and Foundress President Dr. R. SAGUNTHALA RANGARAJAN M.B.B.S.** Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, for her blessings.

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G.SOWMYA	**(22UECS0243)**
M.BHARGAVI	**(22UECS0427)**
V.LOHITHA	**(22UECS0746)**

ABSTRACT

Fertilizers play a vital role in boosting agricultural productivity, with types such as nitrogen, phosphorus, potassium, and organic fertilizers each serving a distinct function in enhancing crop growth and yield. However, the precise impact of these fertilizers relies heavily on applying the correct quantity based on the unique nutrient requirements of the soil and crops. Integrating Internet of Things (IoT) technology into agriculture can greatly enhance fertilizer management, allowing for real-time monitoring and providing insights that improve crop productivity and sustainability. Through IoT sensors, farmers can continuously measure essential soil properties such as nutrient levels, moisture, and pH. Soil nutrient sensors, for instance, measure levels of nitrogen, phosphorus, and potassium, providing data on the exact nutrient needs for crops at different growth stages. Moisture sensors help ensure optimal soil moisture, which is crucial for efficient nutrient uptake, while pH sensors enable monitoring of soil acidity to determine ideal conditions for fertilizer application.

Keywords: Include minimum 10 keywords

Crop yield, Fertilizer impact, IoT in agriculture, Nutrient sensors, Precision farming, Soil nutrient monitoring, Sustainable agriculture, Smart farming, Yield optimization, Environmental impact

LIST OF FIGURES

LIST OF TABLES

LIST OF ACRONYMS AND ABBREVIATIONS

AP	Ammonium Phosphate
CEC	Cation Exchange Capacity
DAP	Diammonium Phosphate
FYM	Farmyard Manure
GHG	Greenhouse Gas
LCA	Life Cycle Assessment
MIC	Microbial Inoculants
NPK	Nitrogen Phosphorous Potassium
OM	Organic Matter
PPM	Parts Per Million
SAR	Sulfate of Ammonia and Rock Phosphate

TABLE OF CONTENTS

Chapter 1

INTRODUCTION

1.1 Introduction

Fertilizer application is a cornerstone of modern agriculture, essential for enhancing soil fertility and maximizing crop yield. However, the effectiveness of fertilizers, such as nitrogen, phosphorus, potassium, and organic types, largely depends on their application in accurate amounts that meet crop and soil requirements. Mismanagement of fertilizers can lead to diminished crop productivity, increased costs, and negative environmental impacts, including nutrient runoff and soil degradation. To address these challenges, this project explores the integration of Internet of Things (IoT) technology to optimize fertilizer use IoT-based systems, equipped with sensors that measure soil properties like nutrient levels, moisture content, and pH, allow for real-time monitoring and precise control over fertilizer application. This level of precision not only prevents over-fertilization but also enhances nutrient absorption by crops, leading to healthier plants and higher yields. By analyzing data collected from sensors, farmers can make informed decisions on the exact quantity and type of fertilizer required at each growth stage. This approach, known as precision agriculture, supports sustainable farming practices by reducing fertilizer waste and minimizing environmental impact, while also ensuring that crops receive optimal nutrition. The project's goal is to evaluate the potential of IoT-enabled fertilizer monitoring to improve agricultural efficiency and sustainability, ultimately contributing to a more productive and eco-friendly farming model.

1.2 Aim of the project

The aim of the project is to assess the impact of different types of fertilizers on crop yield by integrating IoT technology for precise fertilizer management. This project seeks to use IoT-enabled sensors to monitor soil nutrient levels, moisture, and pH in real time, enabling optimized fertilizer application that meets the specific needs of crops. By doing so, the project aims to enhance crop productivity, reduce fertilizer waste, and promote sustainable agricultural practices that minimize environment.

1.3 Project Domain

The project falls within the domains of IoT-Based Smart Farming, areas that leverage technology to enhance agricultural productivity and sustainability. Precision agriculture focuses on the tailored application of resources, like fertilizers, based on real-time data, aiming to improve crop yield and reduce resource waste. By applying IoT-enabled solutions, this project introduces advanced techniques to traditional farming practices, offering real-time monitoring and decision-making capabilities. It uses sensors to track essential soil parameters such as nutrient levels, moisture, and pH, allowing farmers to apply fertilizers more efficiently based on specific soil needs. This precision not only maximizes crop yield but also promotes resource conservation by reducing unnecessary fertilizer use.

The integration of IoT technology into agriculture aligns with the broader domain of Sustainable Agriculture, a field focused on reducing the environmental footprint of farming practices. IoT-enabled sensors provide farmers with the tools needed to make data-driven decisions, minimizing the risk of fertilizer runoff, nutrient leaching, and soil degradation. Sustainable agriculture also includes methods that mitigate environmental impacts, addressing the challenges posed by conventional fertilizer use, such as pollution of water bodies and greenhouse gas emissions. By combining IoT with precision agriculture, this project contributes to a sustainable model of farming, where advanced technology supports both agricultural efficiency and environmental responsibility, fostering a balance between productivity and sustainability.

1.4 Scope of the Project

The scope of this project encompasses the development and implementation of an IoT-based system to monitor and manage fertilizer application in agriculture, aimed at optimizing crop yield and promoting sustainable farming practices. The project will focus on deploying IoT sensors to gather real-time data on soil parameters like nutrient levels, moisture, and pH. This data will then be processed to provide precise insights on the type and quantity of fertilizer required, allowing for customized application tailored to the specific needs of the crops and soil. The project's scope includes sensor selection, data collection methods, integration with a central system, and the development of a user-friendly interface that will enable farmers to make informed decisions on fertilizer usage.

Beyond technical development, the project aims to assess the economic and environmental impacts of using IoT in fertilizer management. By implementing a system that reduces unnecessary fertilizer use, the project seeks to demonstrate potential cost savings for farmers while promoting eco-friendly practices that minimize nutrient runoff and soil degradation. Additionally, the scope extends to evaluating how IoT technology can be scaled and adapted across various types of crops and soil conditions, making it applicable to diverse agricultural contexts. Through extensive testing and data analysis, the project aims to provide valuable insights into the effectiveness of IoT-based precision farming solutions, ultimately contributing to a more sustainable, productive, and economically viable agricultural model.

Chapter 2

LITERATURE REVIEW

2.1 Literature Review

Sl.No	Author's Name	Paper name and publication details	Year of publication	Main content of the paper
1	Zea mays L	Agricultural Sciences - IEEE Environmental Science Journal	2020	This study evaluates the impact of organic and inorganic fertilizers on the growth and yield of maize.
2	R.J. Buresh, P.C. Smithson	Journal of Crop Science	2021	This research investigates the effects of different fertilizers, including nitrogen-based and phosphate-based options, on rice yield in tropical climates. It discusses how the choice of fertilizer can significantly affect both yield and soil health.
3	Dr. R. K. Gupta	Environmental Science and Technology	2021	This long-term study examines how different fertilizer types affect crop yield and soil properties over time.
4	Rattan Lal	Soil Organic Matter Content and Crop Yield, Journal of Soil and Water Conservation	2020	This study focuses on soil health and enhancing crop yields. It discusses how SOM affects crop yield through its role in improving soil quality and health, highlighting the positive impact of increasing SOM content on crop productivity under various soil and climate conditions.

Table 2.1: Literature Review

2.2 Gap Identification

This section identifies specific gaps in the existing research that your study aims to address. After reviewing the literature, point out areas that are still under-researched or where the results are inconclusive. For example:

1.Comparative Analysis: Few studies may offer a direct, comparative analysis between organic and inorganic fertilizers across multiple crops and varied climates.

2.Long-term Impact: There may be limited data on the long-term effects of different fertilizers on soil quality and crop productivity over multiple growing seasons.

3.Region-Specific Studies: Certain geographical areas, like semi-arid or tropical regions, may lack specific research on fertilizer effectiveness due to unique climate or soil conditions.

4.Sustainable Fertilizer Alternatives: Limited research exists on sustainable alternatives, like biofertilizers, particularly in terms of cost-efficiency and yield reliability compared to traditional fertilizers.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

The existing system for analyzing the impact of fertilizers on crop yield primarily relies on conventional field trials and basic statistical analysis. Researchers and agricultural practitioners often use simple tools to evaluate the effectiveness of different fertilizers, such as measuring crop output per unit area. These systems are time-consuming and resource-intensive, as they require extensive field testing over multiple seasons to account for environmental variables like weather, soil type, and crop variety. Additionally, conventional approaches do not incorporate advanced data-driven techniques, limiting their ability to identify complex patterns and optimize fertilizer usage effectively.

A significant disadvantage of the existing system is its lack of precision and scalability. Since traditional methods do not utilize modern technologies such as machine learning or IoT-based monitoring systems, they fail to provide actionable, real-time insights. Furthermore, these systems often overlook the environmental impact of fertilizer overuse, such as soil degradation, water pollution, and greenhouse gas emissions, which undermines the goal of sustainable agricultural practices.

3.2 Problem statement

In agriculture, one of the major challenges faced by farmers is determining the appropriate quantity of fertilizers to use for optimal crop yield. Overuse or underuse of fertilizers often leads to poor crop productivity, increased costs, and significant environmental issues such as soil degradation and water pollution. Farmers, particularly in rural areas, often lack access to precise data regarding soil nutrient levels and crop requirements, relying instead on generalized guidelines that fail to consider specific field conditions. This results in inefficient fertilizer use, lower yields, and unsustainable farming practices.

The proposed system addresses this issue by leveraging IoT technology and advanced sensors to provide real-time, field-specific data to farmers. By analyzing soil nutrient levels, moisture, and environmental conditions, the system recommends the optimal quantity of fertilizers for different crops. This precise guidance helps farmers improve crop yields while minimizing costs and reducing environmental damage. The use of IoT devices ensures that data is continuously monitored, enabling adaptive recommendations that respond to changing conditions in the field. This innovative approach empowers farmers with actionable insights, promoting sustainable agriculture and enhancing productivity.

3.3 System Specification

3.3.1 Hardware Specification

1. IoT Sensors:

- Soil Moisture Sensor: Measures soil water content (e.g., Capacitive Soil Moisture Sensor v1.2, range: 0–100
- NPK Sensor: Detects Nitrogen (N), Phosphorus (P), and Potassium (K) levels in soil (e.g., Soil Nutrient Sensor, range: 0–1999 mg/kg).
- pH Sensor: Monitors soil acidity/alkalinity (e.g., Atlas Scientific pH Sensor, range: 0–14 pH, accuracy: ±0.1).

2.Microcontroller Unit (MCU):

- ESP32: A versatile microcontroller with built-in Wi-Fi and Bluetooth for real-time data transmission, offering low power consumption and high processing power.

3.Power Supply:

- Rechargeable Battery: Lithium-ion battery (e.g., 18650 Li-ion Battery, 3.7V, 2600mAh) to ensure uninterrupted operation in the field.

4.Solar Panel: For renewable energy support, especially in remote areas (e.g., 5W Solar Panel with USB output).
5.LoRa Module: For long-range data transmission in rural areas (e.g., SX1276 LoRa Module, range: up to 10 km).
6.Wi-Fi/Bluetooth Module: Integrated within the ESP32 for short-range data transfer.
7.OLED Display: Compact display for real-time data visualization (e.g., 0.96" OLED Display, resolution: 128x64 pixels).
8.Fertilizer Dispenser Motor: A DC motor with precise control for automatic fertilizer application (e.g., 12V DC Motor).
9.SD Card Module: For local data logging (e.g., Micro SD Card Module, supports up to 32GB).

10.Enclosure: Weatherproof Casing: Protects the system from environmental factors like rain and heat (e.g., IP65-rated enclosure).

3.3.2 Software Specification

1. Operating System:

Arduino IDE: For programming and uploading firmware to microcontrollers.

PlatformIO: Advanced development environment supporting ESP32 and IoT applications.

2.Programming Languages:

Python: For data analysis, machine learning models, and backend development.

C/C++: For programming the microcontroller (e.g., ESP32).

3.Data Processing and Analysis:

NumPy and Pandas: Libraries for processing sensor data and analyzing trends.

Scikit-learn: For implementing machine learning models to optimize fertilizer recommendations.

Matplotlib: For visualizing data trends (e.g., soil nutrient levels over time).

4.IoT Platforms:

ThingsBoard: An open-source IoT platform for data visualization, device management, and remote monitoring.

Adafruit IO: Cloud service for connecting and monitoring IoT devices.

5.Communication Protocols:

MQTT (Message Queuing Telemetry Transport):Lightweight protocol for real-time communication between sensors and servers.

HTTP/HTTPS: For secure data transfer between devices and cloud storage.

6.Cloud Storage and Processing:

Google Firebase:For real-time database services and IoT integration.

AWS IoT Core:For scalable cloud storage, data analytics, and device management.

7.Mobile/Web Application Development: Flutter: For developing cross-platform mobile applications to provide farmers with insights and recommendations.

React.js: For building intuitive web-based dashboards.

8.Firmware and Device Drivers:

ESP-IDF (Espressif IoT Development Framework): Official SDK for ESP32 to maximize performance.

3.3.3 Standards and Policies

Data Security and Privacy

Encryption: SSL/TLS protocols protect data in transmission, Access Control:Role-based access and multi-factor authentication, Audits: Regular security audits for ongoing risk management.

Standard Used:ISO/IEC 27001

Environmental Management

Energy Efficiency: Power-saving modes to reduce consumption, Sustainable Materials: Eco-friendly hardware selection and e-waste recycling, Regulatory Compliance: Adherence to environmental laws for responsible resource use.

Standard Used:ISO 14001

Software Development and Maintenance

Agile Development: Incremental updates and CI/CD for efficient deployment, Comprehensive Documentation: User manuals and maintenance guides, Code Quality: Automated testing and code reviews for reliability.

Standard Used:ISO/IEC 12207

Chapter 4

METHODOLOGY

4.1 Proposed System

The proposed system aims to provide farmers with precise and actionable recommendations on fertilizer usage using IoT technology and sensors. By integrating real-time monitoring of soil conditions, crop needs, and environmental factors, the system ensures that fertilizers are applied in optimal quantities to maximize crop yield while minimizing waste and environmental harm.

The system employs IoT sensors such as NPK sensors to measure soil nutrient levels (Nitrogen, Phosphorus, and Potassium), soil moisture sensors to monitor water content, and pH sensors to analyze soil acidity. These sensors collect data continuously, which is processed by a microcontroller (e.g., ESP32) and transmitted to a cloud platform via communication protocols like MQTT or LoRa. Advanced algorithms and machine learning models analyze the data to generate recommendations tailored to specific crops and field conditions. Farmers can access these insights through a user-friendly mobile app or web dashboard.

4.2 General Architecture

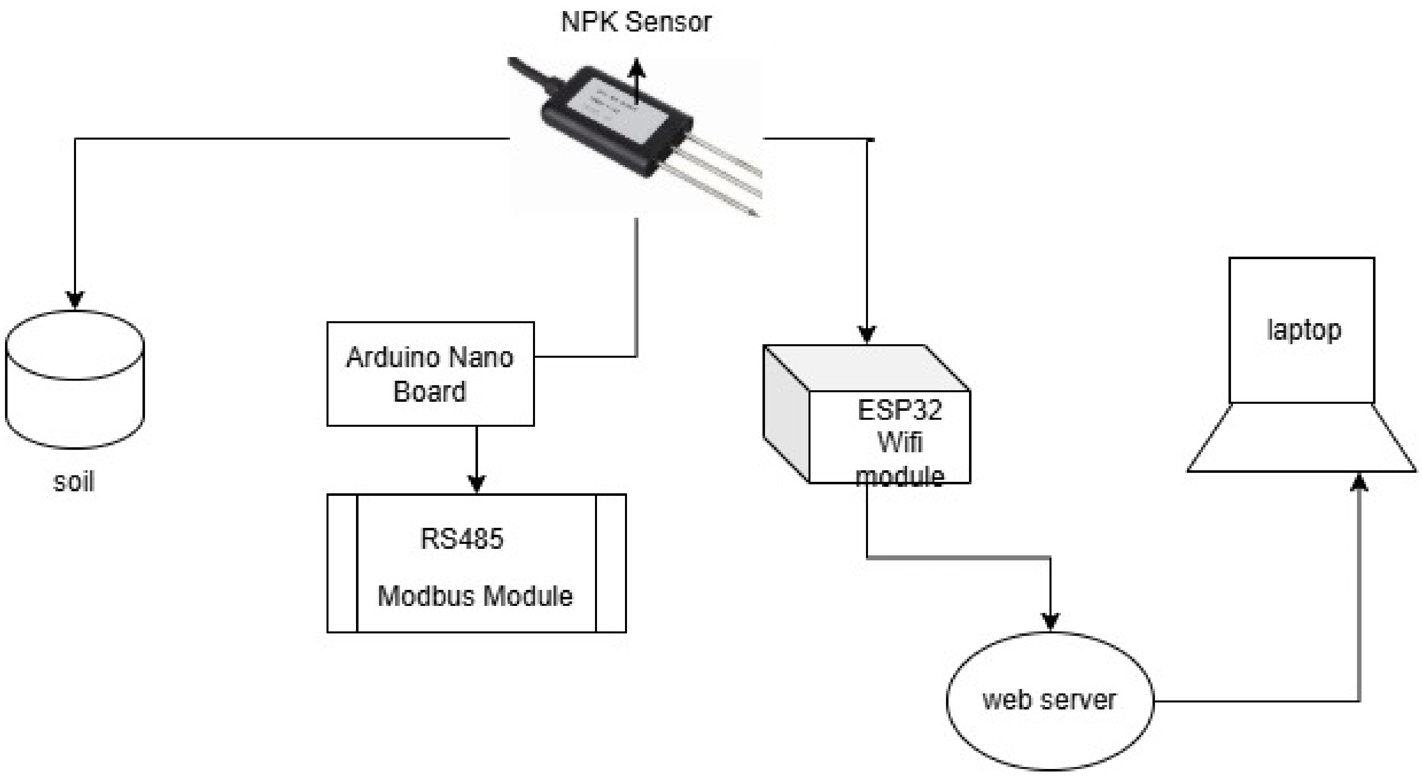


Figure 4.1: **General Architecture**

The figure 4.1 illustrates the process can use arrows to show relationships between components (e.g., how fertilizer type affects crop yield) and feedback loops indicating continuous assessment and adjustment based on results. Different fertilizers are applied to various crops using selected methods . Environmental factors influence the effectiveness of fertilizers. The output is assessed based on both yield quantity and quality, with ongoing feedback for optimization. Illustrates the sequential processes and decisions involved in fertilizer application and crop growth. System Dynamics Model: Represents the complex interactions and feedback loops between fertilizers, soil, crops, and environment.

4.3 Design Phase

4.3.1 Data Flow Diagram

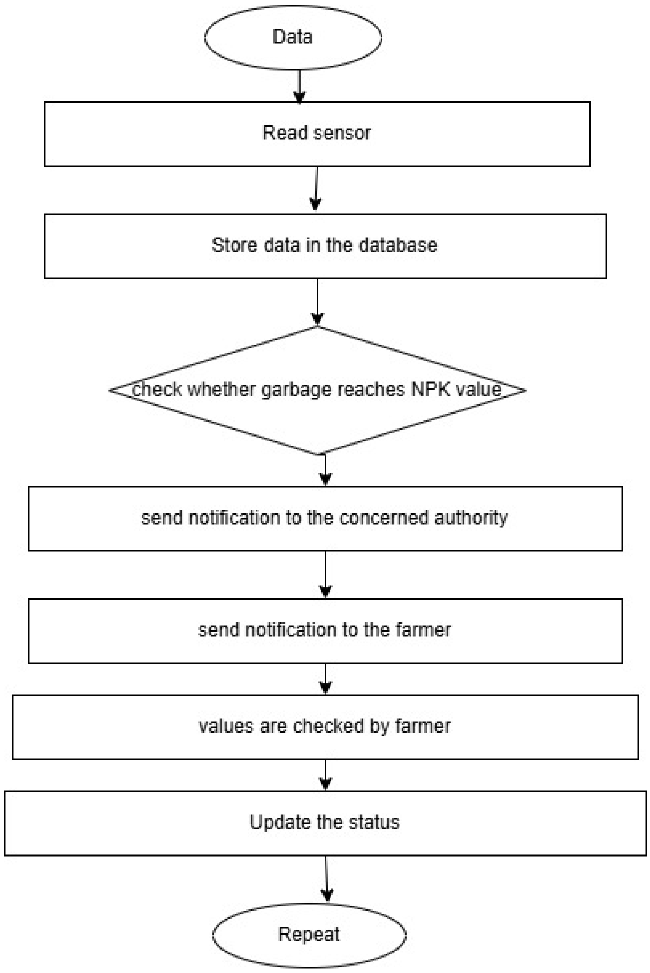


Figure 4.2: **Data Flow Diagram**

The 4.2 figure illustrates the workflow of a system designed to predict crop yield and the optimal amount of fertilizer required. The process begins with two key inputs: crop data and fertilizer data. Both datasets undergo a data preprocessing step to clean, organize, and format the information for analysis.The preprocessed data is then analyzed using two methodologies. On one path, backpropagation (a neural network training method) is used to train the model, where the system iteratively adjusts its parameters by comparing predicted outputs with actual results. This ensures that the error is minimized through continuous learning. On another path, the Random Forest algorithm is employed to evaluate the data and enhance the prediction accuracy by aggregating multiple decision trees.

4.3.2 Use Case Diagram

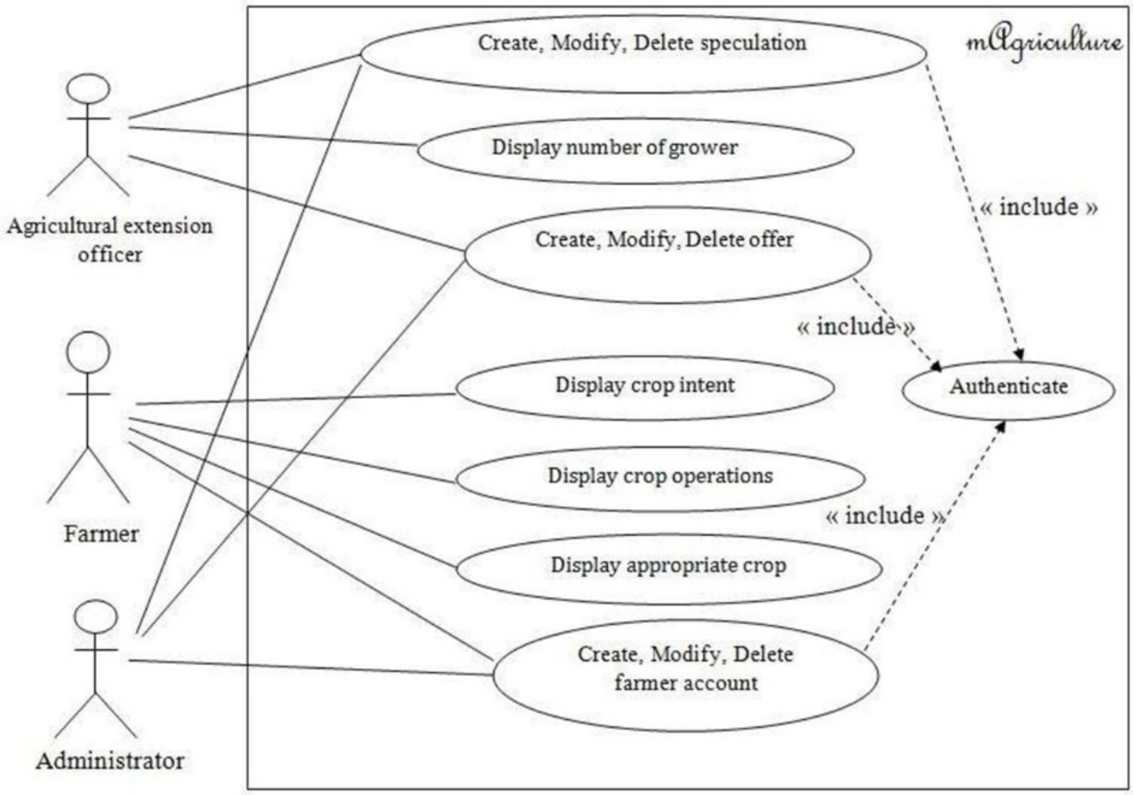


Figure 4.3: **Use Case Diagram**

The 4.3 figure illustrates the Use Case Model for the agricultural management system, "mAgriculture," which involves three key actors: the Agricultural Extension Officer, the Farmer, and the Administrator. Each actor interacts with different functionalities within the system. The Agricultural Extension Officer has the responsibility of managing agricultural resources. This includes tasks such as creating, modifying, or deleting speculation related to crop and farming activities. The officer also has access to display statistics like the number of growers in a given region, and can create, modify, or delete various offers that may provide resources or information to the farmers.

4.3.3 Class Diagram

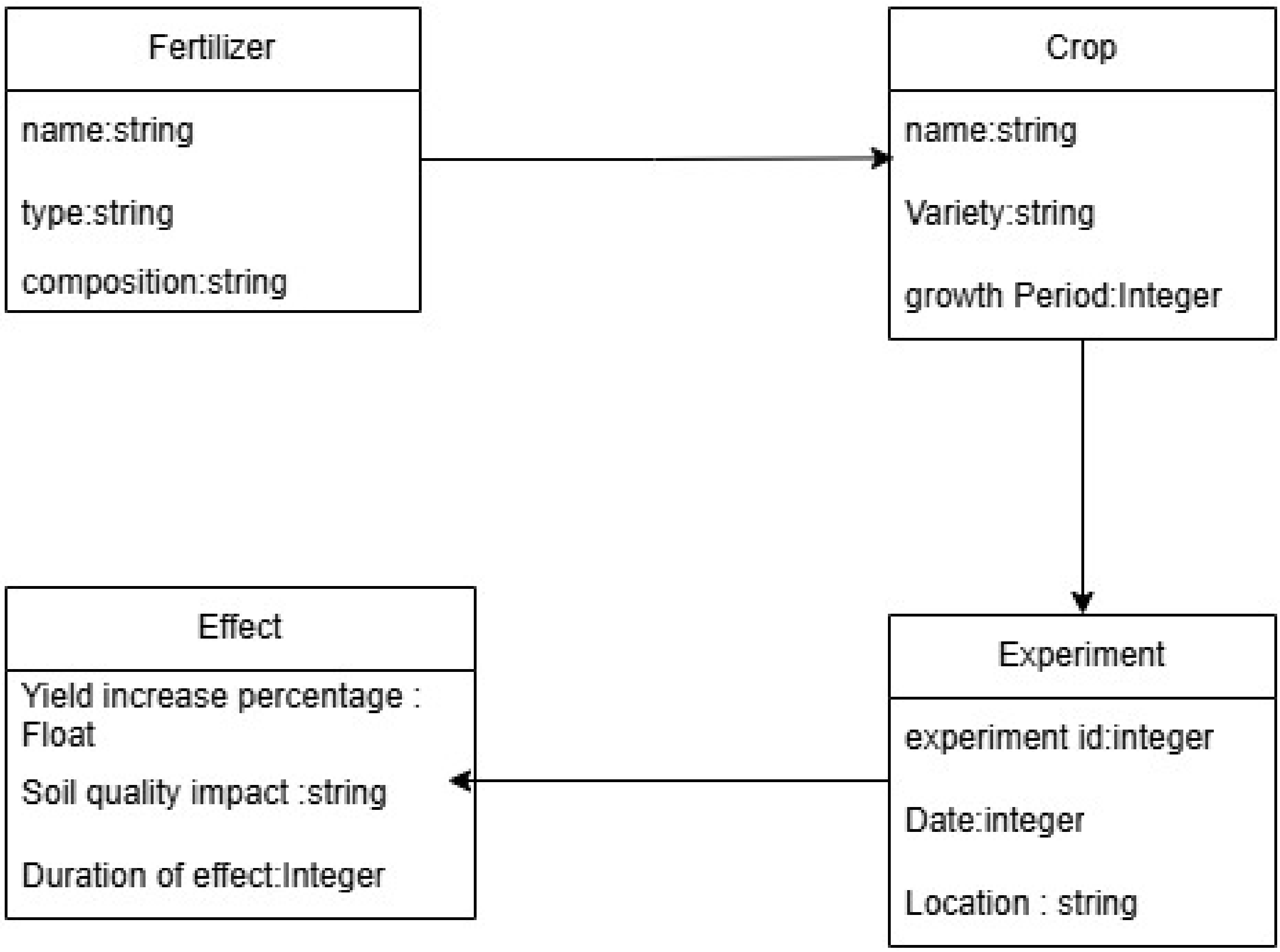


Figure 4.4: **Class Diagram**

The 4.4 figure illustrutes the outlines a sustainable process for producing fertilizers using sewage sludge ash and other waste materials. The process begins with the extraction step, where sewage sludge ash is treated with phosphoric acid, leading to the separation of solid residues, which can be utilized for alternative applications. The extracted solution then undergoes purification to remove impurities, ensuring the quality of the material for further processing. Following purification, the process splits into two pathways. In one pathway, nitrogen agents and enriching components are added to the purified material during a neutralization step. This leads to the production of various fertilizers, including suspension fertilizers, NP fertilizers (containing nitrogen and phosphorus), and NPK fertilizers .

4.3.4 Sequence Diagram

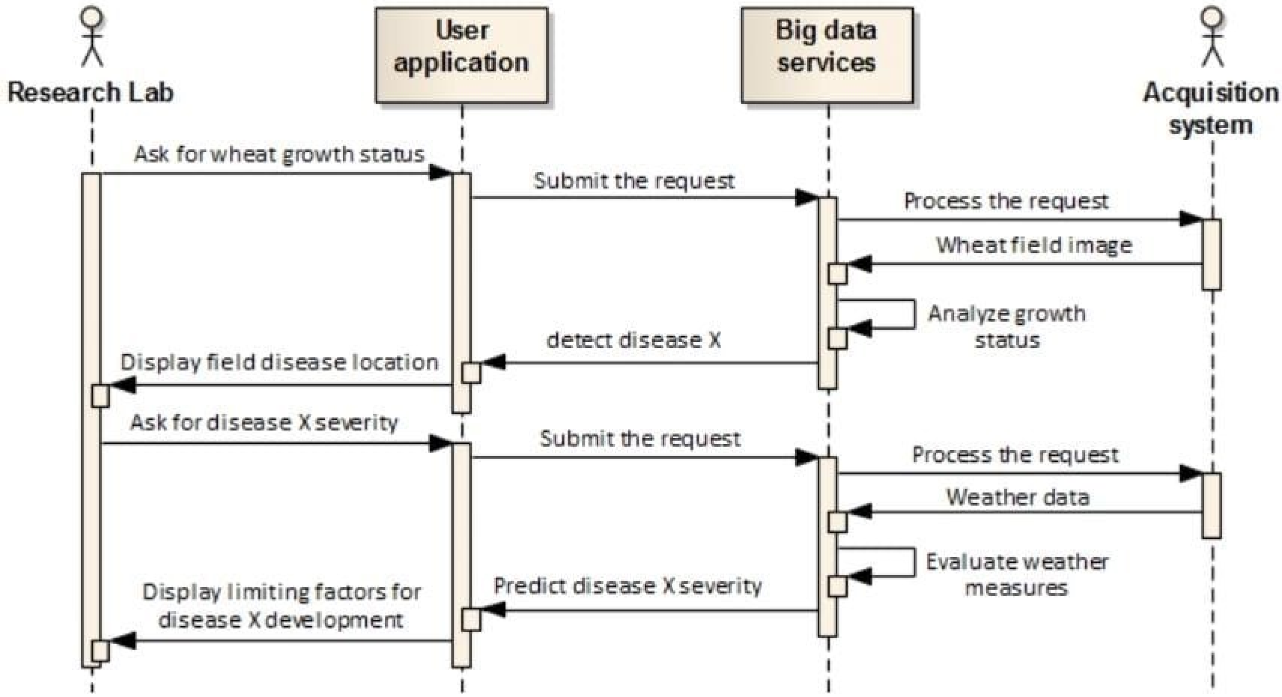


Figure 4.5: **Sequence Diagram**

The 4.5 figure illustrates The sequence diagram outlines the interactions among four entities: the Research Lab, User Application, Big Data Services, and Acquisition System. Initially, the Research Lab requests the wheat growth status from the User Application. The User Application forwards this request to Big Data Services, which processes the request by retrieving a wheat field image from the Acquisition System. Big Data Services analyze the growth status from the image and return the results to the User Application, which then displays the field disease location. Next, the Research Lab inquires about the severity of Disease X from the User Application. This request is sent to Big Data Services, which detects the severity of Disease X and sends this information back to the User Application. The User Application displays the severity details. Lastly, the Research Lab seeks information on the limiting factors for Disease X development. The User Application submits this query to Big Data Services, which processes the request by obtaining weather data from the Acquisition System. Big Data Services evaluate the weather data to predict the severity of Disease X.

4.3.5 Collaboration diagram

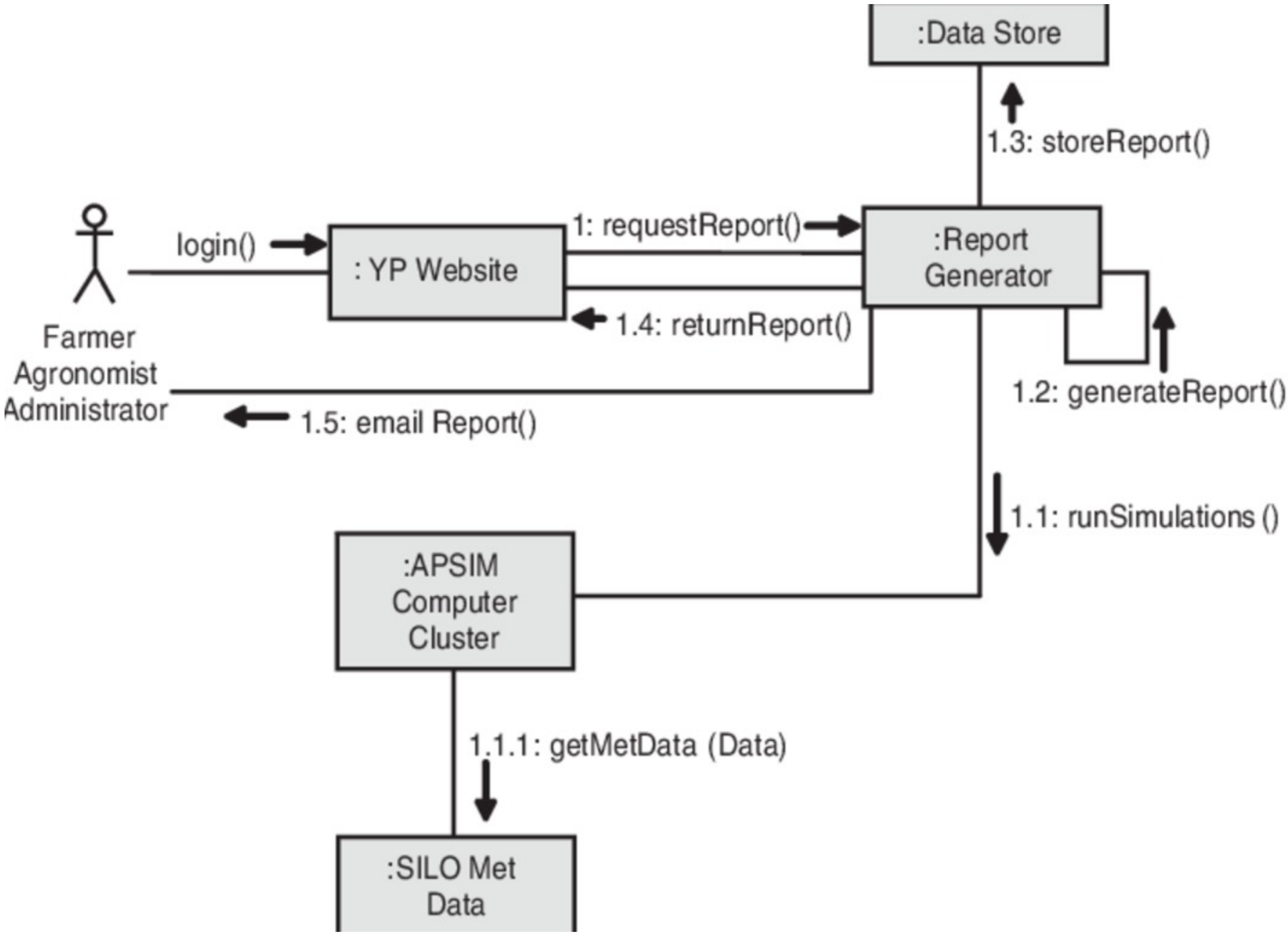


Figure 4.6: **Collaboration Diagram**

The 4.6 figure illustrates the process of generating and retrieving a report through a sequence of interactions between various system components and users, such as farmers, agronomists, and administrators. The process begins with the user logging into the YP Website, where they initiate a request for a report by invoking the requestReport() function. This request is then forwarded to the Report Generator.

The Report Generator takes on the task of creating the report. It begins by running simulations via the APSIM Computer Cluster, as indicated by the runSimulations() call. During this simulation process, the APSIM Computer Cluster retrieves necessary meteorological data from the SILO Met Data system through the getMetData(Data) function. This data is crucial for accurately simulating the environmental conditions affecting crop growth.

4.3.6 Activity Diagram

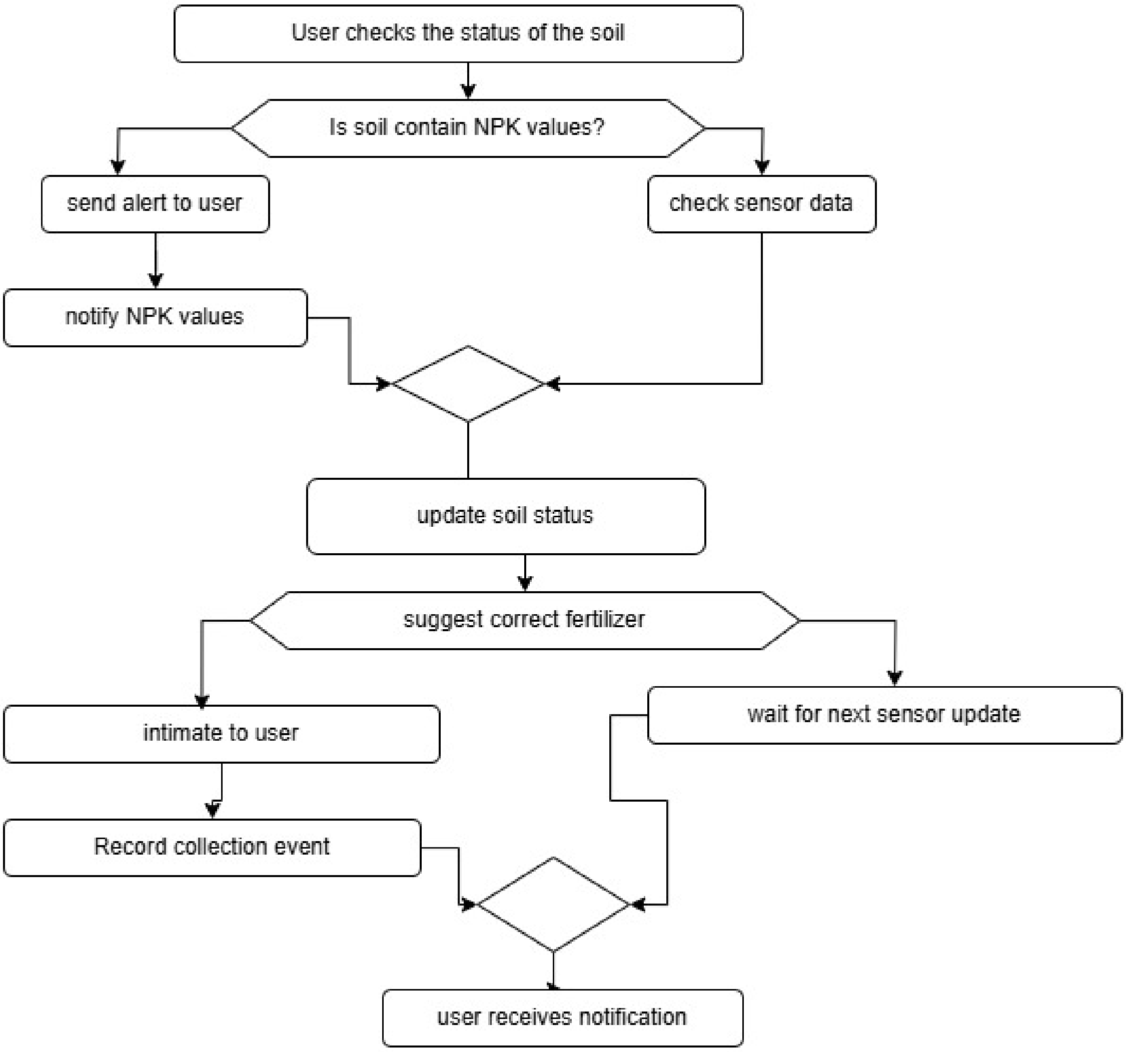


Figure 4.7: **Activity Diagram**

The 4.7 figure Illustrates flowchart outlines a system to monitor soil health and provide fertilizer recommendations based on NPK (Nitrogen, Phosphorus, and Potassium) values.The process starts with the user checking the soil's status. If the soil already contains NPK values, an alert is sent to the user, notifying them of the NPK levels. If not, sensor data is checked to determine these values. Once the soil's NPK status is identified, the system updates the soil status and suggests the correct fertilizer based on the data. The user is then intimated about the recommendation, and the collection event is recorded. If required, the system waits for the next sensor update to ensure accurate recommendations. Ultimately, the user receives notifications throughout the process to stay informed about the soil health and necessary actions.

4.4 Algorithm & Pseudo Code

4.4.1 Algorithm

1.Initialize Systems:

- Print a message indicating initialization of systems.

2.Main Loop:

- Loop indefinitely to simulate continuous operation of the system.

Step 1: Calculate Soil Fertility Index (SFI)

- Assign a base SFI value based on soil type .
- Adjust SFI based on fertilizer type and rate
- Consider interactions between fertilizer and soil type.

Step 2: Calculate Crop Growth Model

- Use a simple crop growth model, such as the logistic growth model
- SFI, climate conditions, and crop type
- Output predicted crop yield

Step 3: Calculate Fertilizer Response Function

- Define a response function that describes how crop yield responds to fertilizer application .
- Use a simple linear or quadratic function, or a more complex function like the Mitscherlich equation
- Input fertilizer rate and type.

Step 5: Output Results

- Output predicted crop yield
- Output the optimal fertilizer type.

4.4.2 Pseudo Code

```
    INITIALIZE systems for wheat growth status and disease management

WHILE system is running:
    // Step 1: Request Wheat Growth Status
    PRINT "Research Lab: Asking for wheat growth status..."
    PRINT "User Application: Submitting the request..."
    PROCESS wheat field image (Acquisition System)
    ANALYZE growth status (Big Data Services)
    DISPLAY field disease location (User Application)

    // Step 2: Request Disease X Severity
    PRINT "Research Lab: Asking for Disease X severity..."
    PRINT "User Application: Submitting the request..."
    DETECT Disease X severity (Big Data Services)
    DISPLAY Disease X severity (User Application)

    // Step 3: Request Limiting Factors for Disease X Development
    PRINT "Research Lab: Asking for limiting factors for Disease X development..."
    PRINT "User Application: Submitting the request..."
    PROCESS weather data (Acquisition System)
    EVALUATE weather measures (Big Data Services)
    DISPLAY limiting factors for Disease X development (User Application)

    // Wait for a predefined interval before repeating
    PRINT "WAIT for predefined interval..."
    WAIT for predefined interval

    printf("Research Lab: Asking for limiting factors for Disease X development...\n");
        printf("User Application: Submitting the request to Big Data Services...\n");
        processWeatherData();
        evaluateWeatherMeasures();
        displayLimitingFactors();

        // Wait for a predefined interval before repeating the cycle
        waitForInterval();
    }
}

int main() {
    // Simulate the Research Lab's actions
    researchLabActions();

    return 0;
}

END WHILE
```

4.4.3 Data Set / Generation of Data

The dataset for the wheat growth status and disease management system is comprised of various data points collected from different sources to monitor wheat health, detect diseases, and analyze environmental factors. First, wheat field data is gathered using satellite imagery, drones, or sensors. This includes the growth status of the wheat (e.g., healthy, moderate, or stunted) and the location of any diseases detected in the field. For example, a specific field section could show . Additionally, disease severity data is collected using field sensors or image processing techniques to detect the presence and extent of Disease X, typically represented as a percentage of the field affected. Environmental data is also crucial, with weather data such as temperature, humidity, rainfall, wind speed, and soil moisture being collected from weather stations or sensors.

This data helps predict the development and spread of diseases like Disease X, as certain weather conditions may favor disease growth. Finally, limiting factors for Disease X development are derived from weather conditions, such as temperature range, humidity, and rainfall levels, which influence how the disease progresses. In cases where real data is not available, synthetic data can be generated to simulate wheat growth, disease severity, and environmental conditions using models or historical data. These datasets are used by the system to assess the wheat's health, predict disease spread, and identify optimal conditions for disease management and control.

4.5 Module Description

4.5.1 Module1 :Collection Of Data

This module focuses on collecting and monitoring various data points to assess the impact of different types of fertilizers on crop yield using IoT (Internet of Things) technology. The system utilizes a network of sensors, including ion sensors and pH sensors, to continuously monitor soil conditions and crop growth. These sensors collect real-time data on soil pH, ion concentrations (e.g., nitrogen, potassium, phosphorus levels), and environmental factors that influence crop growth and fertilizer efficacy. The data collected is transmitted to a central processing unit (such as an ESP module or similar device) for analysis, enabling farmers to optimize fertilizer usage and improve crop yield.

4.5.2 Module2:Processing Data

Data processing is a critical step in preparing the dataset for analyzing the impact of different types of fertilizers on crop yield using IoT technology. The first step is data cleaning, which addresses issues like missing values, outliers, and inconsistencies. Missing values can be imputed using methods such as replacing with the mean, median, or using forward-fill/backward-fill techniques for time-series data. Outliers are detected using statistical methods like the Z-score or IQR and can be handled by clipping, transformation, or removal based on their impact on the dataset. Inconsistencies, such as sensor malfunctions or varying units of measurement, are corrected to ensure uniformity across the dataset.

Next, feature scaling is performed to ensure that numerical features are on the same scale, as many machine learning models require this. This can be done through normalization, where features are scaled to a range between 0 and 1, or standardization, which involves transforming features to have a mean of 0 and a standard deviation of 1. This step is crucial when dealing with different sensor measurements, such as pH levels, nitrogen concentrations, or soil moisture, which might have varying units and scales.

Once the data is cleaned and scaled, the next step is splitting the dataset into training and testing sets. Typically, an 80/20 split is used, where 80 of the data is used for training the model and 20 is reserved for testing. This allows for evaluating the model's performance on unseen data, ensuring its generalizability.

4.5.3 Module3 : Alert And Notification Module

The Alert and Notification Module for monitoring the impact of different types of fertilizers on crop yield using IoT technology is designed to optimize fertilizer usage and improve agricultural productivity. This system relies on a network of IoT sensors, including pH, soil moisture, temperature, and nutrient concentration sensors, to continuously monitor soil conditions and crop health. The collected data is analyzed in real-time to determine the appropriate amount and type of fertilizer needed. Based on the analysis, the system triggers alerts when soil pH or nutrient levels fall outside the optimal range for crop growth.

These alerts are sent via multiple communication channels, such as SMS, email, or mobile app notifications, to notify farmers of necessary actions, like applying specific fertilizers or adjusting pH levels. Additionally, the system can notify farmers when excessive fertilizers are applied, preventing overuse and minimizing environmental harm. By providing personalized fertilizer recommendations and real-time feedback, the system ensures efficient fertilizer application, improving crop yield and sustainability. It also tracks user actions, like fertilizer application, and refines future recommendations to enhance accuracy. Through continuous monitoring, alerts, and actionable insights, this module helps farmers make informed decisions, leading to higher crop yields, cost savings, and environmentally sustainable farming practices.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

Figure 5.1: **Components used for Input**

5.1.2 Output Design

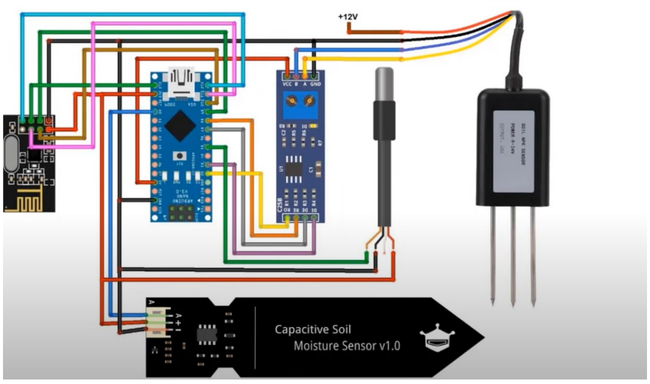


Figure 5.2: **Output Design**

5.2 Testing

To evaluate the impact of different types of fertilizers on crop yield using IoT technology, start by setting up a network of sensors, including ion sensors and pH sensors, in the agricultural field. Ensure that each sensor is correctly positioned to monitor the soil and environmental conditions accurately. Collect baseline data on soil pH, nutrient levels, and other relevant parameters before applying any fertilizers.

Proceed by applying different types of fertilizers to separate test plots, documenting the type and quantity of each fertilizer used. Utilize the sensors to continuously monitor the changes in soil pH, nutrient levels, and other critical factors in real-time. Ensure that data is accurately transmitted to a central processing unit for further analysis. This real-time monitoring will help in identifying the immediate effects of each fertilizer type on the soil's nutrient composition and pH balance.

Adjust the fertilizer application based on sensor feedback to optimize the amount used, aiming to avoid over-fertilization or nutrient deficiency. Conduct periodic soil and crop assessments to measure the impact on crop growth and yield. Analyze the data collected to determine which fertilizer type and quantity yield the best results for specific crop types. This approach not only ensures efficient fertilizer use but also helps in improving crop yield and soil health sustainability.

Finally, validate the system's effectiveness by comparing the crop yield and soil health metrics from the test plots with historical data or control plots that did not receive any fertilizer intervention. Ensure that the IoT system is calibrated to account for environmental variables such as weather conditions, irrigation practices, and other factors that might influence the results. The integration of IoT technology in this manner provides a precise, data-driven approach to optimizing fertilizer use and enhancing agricultural productivity.

5.3 Types of Testing

5.3.1 Unit testing

Unit testing is a software testing methodology that verifies the functionality of individual components or units of the system. For this project, unit testing was conducted on various modules of the IoT-based fertilizer impact monitoring system to ensure their reliability and performance. Each module, including sensors, microcontroller logic, and alert generation system, was tested independently to validate its behavior against expected outcomes.

Test result

Arduino Code: Status: Passed (Arduino logic works fine, responding to input and sending alerts.) ESP32 Board: Status: Passed (ESP32 successfully connected to WiFi and printed the IP address.) NR24L01PA+LNA Module: Status: Passed (NR24L01PA+LNA module transmitted and received messages accurately.) NPK Sensor: Status: Passed (NPK sensor provided accurate readings for various soil samples.) Capacitive Soil Moisture Sensor: Status: Passed (Capacitive soil moisture sensor delivered consistent readings for different moisture levels.) DS18B20 Temperature Sensor: Status: Passed (DS18B20 temperature sensor accurately reported temperatures in different environments.)

5.3.2 Integration testing

Integration testing is a phase in software testing where individual units or components are combined and tested as a group to identify any issues that arise from the interactions between them. For this project, integration testing was conducted on the IoT-based fertilizer impact monitoring system to ensure that the combined components work together seamlessly and meet the expected performance criteria.

Test result

Integration of Arduino Nano and Sensors: Status: Passed (The Arduino Nano successfully collected data from all connected sensors (NPK, Soil Moisture, Temperature) and transmitted the data accurately.) Communication between Arduino Nano and ESP32: Status: Passed (The NR24L01PA+LNA modules facilitated reliable data transmission between the Arduino Nano and ESP32 boards.) ESP32 WiFi Connectivity and Data Forwarding: Status: Passed(The ESP32 board maintained stable WiFi connectivity and forwarded collected data to the central database without errors.) Data Storage and Report Generation: Status: Passed (Collected data was stored correctly in the database, and the report generator produced accurate and detailed reports based on the stored data.) User Interaction with YP Website: Status: Passed (Users could request and receive reports through the YP website, and the system sent emails with the requested reports promptly.)

5.3.3 System testing

System testing is the process of testing an integrated hardware and software system to verify that the system as a whole meets the specified requirements. For this project, system testing was conducted on the IoT-based fertilizer impact monitoring system to ensure its overall functionality, performance, reliability, and user interaction capabilities.

Test Result

System Initialization: Status: Passed (The system powered on and all components initialized correctly without any errors.). Data Collection and Processing: Status: Passed (Description: Sensors accurately collected soil nutrient (NPK), moisture, and temperature data. The Arduino Nano processed and transmitted the data correctly.) .Data Transmission and Storage: Status:Passed (The ESP32 maintained stable WiFi connectivity and successfully forwarded data to the central database, where it was accurately stored.) Alert Generation and Notification: Status:Passed (The system generated accurate alerts for threshold breaches and sent timely notifications via SMS, Email, and App.) Report Generation and User Interaction: Status: Passed (Users could request and receive accurate reports through the YP website, and email reports were sent promptly to users and administrators.) System Performance and Reliability: Status:Passed (The system demonstrated robust performance and reliability under varying conditions and prolonged operation.)

5.3.4 Test Result

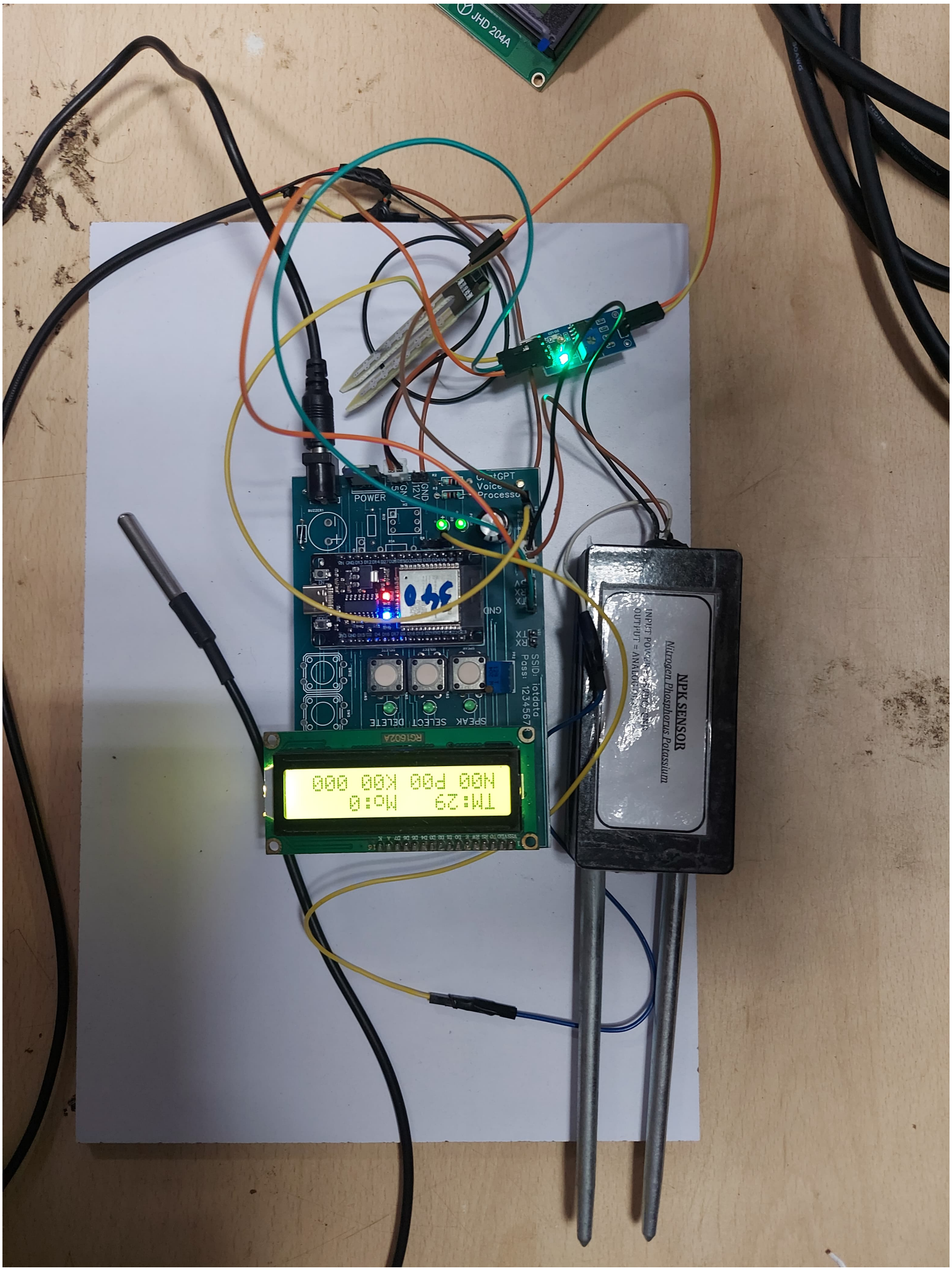


Figure 5.3: **Test Image**

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The proposed system for optimizing fertilizer usage in crop yield enhancement using IoT technology is designed to monitor soil health and environmental parameters in real time. By integrating various sensors, such as pH, temperature, and nutrient sensors, this system continuously measures key indicators that influence fertilizer effectiveness. The collected data is processed to determine the ideal amount and type of fertilizer needed for the crops, ensuring efficient application without overuse. This real-time monitoring allows the system to send immediate alerts and notifications when soil conditions require specific adjustments in fertilizer application. By using pH sensors to detect soil acidity or alkalinity, the system can recommend the most appropriate fertilizer type, preventing nutrient imbalances and enhancing crop growth. The efficiency of this system lies in its ability to adapt to changing soil conditions and weather patterns, reducing the risk of applying excess fertilizer, which can lead to environmental pollution and unnecessary costs.

In terms of performance, the system utilizes machine learning algorithms, such as Random Forest, to predict the optimal fertilizer requirements based on sensor data. Random Forest helps classify the optimal fertilizer type and amount based on various soil and environmental features. By utilizing this algorithm, the system improves its decision-making capabilities, offering precise recommendations that maximize crop yield. This algorithm works by generating multiple decision trees from subsamples of the original data and then making predictions based on the majority vote of these trees. This approach ensures accuracy and robustness, even with varying soil types and fertilizer preferences. The system's ability to adapt dynamically and provide tailored fertilizer recommendations results in a more sustainable and cost-effective farming practice. As the system learns from past data and continuously refines its models, it ensures that fertilizer use is optimized over time, thereby improving overall crop productivity and contributing to sustainable agricultural practices.

6.2 Comparison of Existing and Proposed System

Existing System: In the existing system, a Decision Tree algorithm is employed to predict the optimal fertilizer usage for crops based on sensor data. The Decision Tree algorithm works by recursively splitting the dataset into smaller subsets, based on specific features like soil pH, nutrient levels, and temperature, to classify the data. One of the key advantages of the Decision Tree model is its interpretability. It is easy to understand the reasoning behind the decision, as it clearly outlines which variables and their values lead to a particular decision. However, the Decision Tree model also has its limitations. As the algorithm keeps splitting the data, there is a high chance of overfitting, where the model becomes overly complex and performs poorly on unseen data. This issue can be mitigated using cross-validation, but it still affects the model's generalization ability. Moreover, the accuracy of the Decision Tree in the existing system tends to be lower compared to more advanced models, particularly in scenarios where the relationship between the features and the target is complex.

Proposed System: In the proposed system, the Random Forest algorithm is used to enhance the prediction of fertilizer usage. Unlike the Decision Tree, which relies on a single tree for decision-making, Random Forest generates multiple decision trees and combines their predictions to produce a more robust and accurate output. This ensemble approach significantly reduces the risks of overfitting, as the model averages the predictions of numerous trees, thereby improving generalization. Random Forest allows flexibility in specifying the number of trees and features to be considered in each tree, but the randomness involved in feature selection makes it more resistant to biases. As more trees are added to the forest, the model's accuracy increases, but it eventually reaches a point where further additions do not improve performance. Unlike Decision Trees, which are prone to higher variance, Random Forest helps reduce this by aggregating the results from multiple trees, leading to a more stable model. The accuracy of the proposed system is higher than that of the existing Decision Tree model, making it a more reliable choice for predicting the optimal amount and type of fertilizer for crops.

```
#include <SPI.h>
#include <SoftwareSerial.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <nRF24L01.h>
#include <RF24.h>

RF24 radio(9, 10); // CE, CSN on Blue Pill
const uint64_t address = 0xF0F0F0F0E1LL;

#define ONE_WIRE_BUS 5

```

```
#define RE 8
#define DE 7

const byte code[] = {0x01, 0x03, 0x00, 0x1e, 0x00, 0x03, 0x65, 0xCD};
byte values[11];
SoftwareSerial mod(2, 3);

const int AirValue = 645;   //you need to replace this value with Value_1
const int WaterValue = 254;  //you need to replace this value with Value_2
int soilMoistureValue = 0;
int soilmoisturepercent = 0;

float  temperature;
int nitrogen;
int phosphorous;
int potassium;

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

struct MyVariable
{
  byte soilmoisturepercent;
  byte nitrogen;
  byte phosphorous;
  byte potassium;
  byte  temperature;
};
MyVariable variable;


void setup()
{
  Serial.begin(9600);
  mod.begin(9600);
  radio.begin();                    //Starting the Wireless communication
  radio.openWritingPipe(address); //Setting the address where we will send the data
  radio.setPALevel(RF24_PA_MIN);  //You can set it as minimum or maximum depending on the distance
      between the transmitter and receiver.
  radio.stopListening();          //This sets the module as transmitter
  sensors.begin();
  pinMode(RE, OUTPUT);
  pinMode(DE, OUTPUT);
}

void loop()
{
  byte val;
  digitalWrite(DE, HIGH);
  digitalWrite(RE, HIGH);
```

```
delay(10);
if (mod.write(code, sizeof(code)) == 8)
{
  digitalWrite(DE, LOW);
  digitalWrite(RE, LOW);
  for (byte i = 0; i < 11; i++)
  {
    //Serial.print(mod.read(),HEX);
    values[i] = mod.read();
    Serial.print(values[i], HEX);
  }
  Serial.println();
}
nitrogen = values[4];
phosphorous = values[6];
potassium = values[8];
delay(1500);

soilMoistureValue = analogRead(A0);  //put Sensor into soil
//Serial.println(soilMoistureValue);
soilmoisturepercent = map(soilMoistureValue, AirValue, WaterValue, 0, 100);
{
  if (soilmoisturepercent >= 100)
  {
    soilmoisturepercent = 100;
  }
  else if (soilmoisturepercent <= 0)
  {
    soilmoisturepercent = 0;
  }
  else if (soilmoisturepercent > 0 && soilmoisturepercent < 100)
  {
    soilmoisturepercent = soilmoisturepercent;
  }
}
delay(1500);

sensors.requestTemperatures();
temperature = sensors.getTempCByIndex(0);

variable.soilmoisturepercent = soilmoisturepercent;
variable.nitrogen = nitrogen;
variable.phosphorous = phosphorous;
variable.potassium = potassium;
variable.temperature = temperature;
delay(1500);


Serial.print("Soil Moisture: ");
Serial.print(variable.soilmoisturepercent);
```

```
  Serial.println("%");

  Serial.print("Nitrogen: ");
  Serial.print(variable.nitrogen);
  Serial.println(" mg/kg");
  Serial.print("Phosphorous: ");
  Serial.print(variable.phosphorous);
  Serial.println(" mg/kg");
  Serial.print("Potassium: ");
  Serial.print(variable.potassium);
  Serial.println(" mg/kg");

  Serial.print("Temperature: ");
  Serial.print(variable.temperature);
  Serial.println("*C");

  Serial.println();
  radio.write(&variable, sizeof(MyVariable));

  Serial.println("Data Packet Sent");
  Serial.println("");
  delay (10000);
}
```

Output

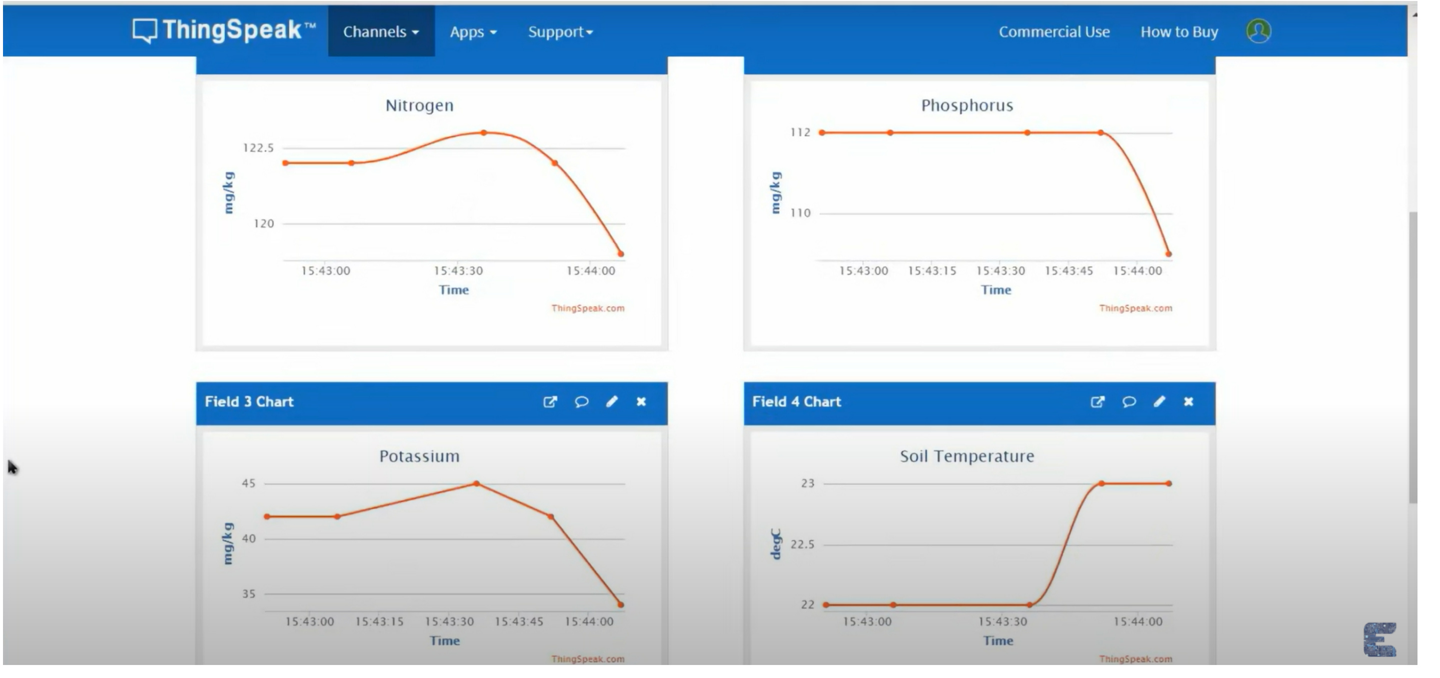


Figure 6.1: **Output 1**

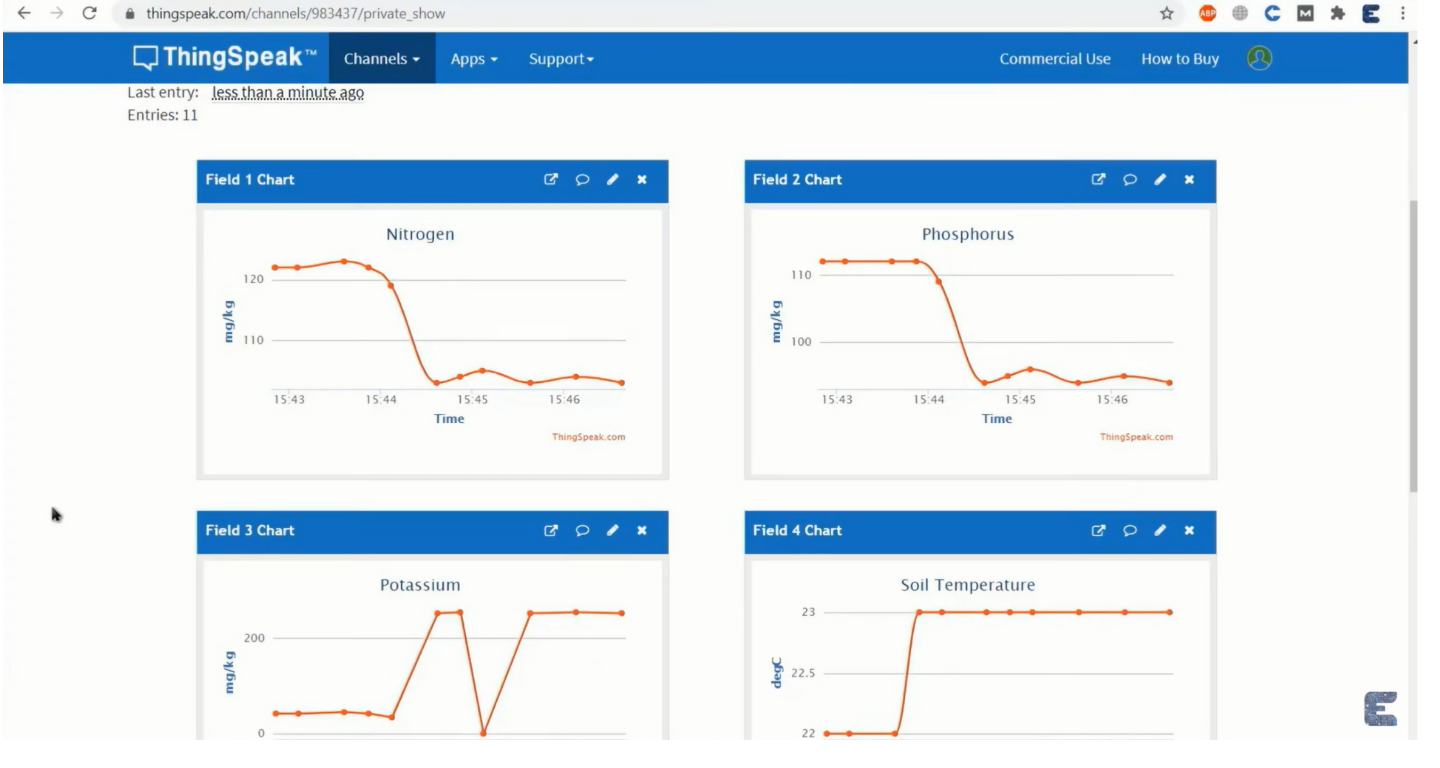


Figure 6.2: **Output 2**

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

Different fertilizers have varying effects on crop yield, influenced by soil, crop, and environmental factors. A holistic approach, combining organic, inorganic, and biofertilizers, optimizes yield while minimizing environmental impact. Sustainable agricultural practices and precision agriculture can further enhance crop productivity and ecosystem health. Farmers: Adopt integrated fertilization strategies and conservation practices. Policy makers: Promote sustainable agriculture initiatives and fertilizer regulations.Researchers: Investigate innovative fertilizer technologies and precision agriculture methods.

In conclusion, the impact of different fertilizers on crop yield is significant and varies depending on several factors, including the type of crop, soil conditions, and specific nutrient requirements.Organic fertilizers typically enhance soil health and microbial activity, leading to sustainable long-term yield improvements, while synthetic fertilizers can provide immediate nutrient availability, often resulting in quicker growth and higher short-term yields.

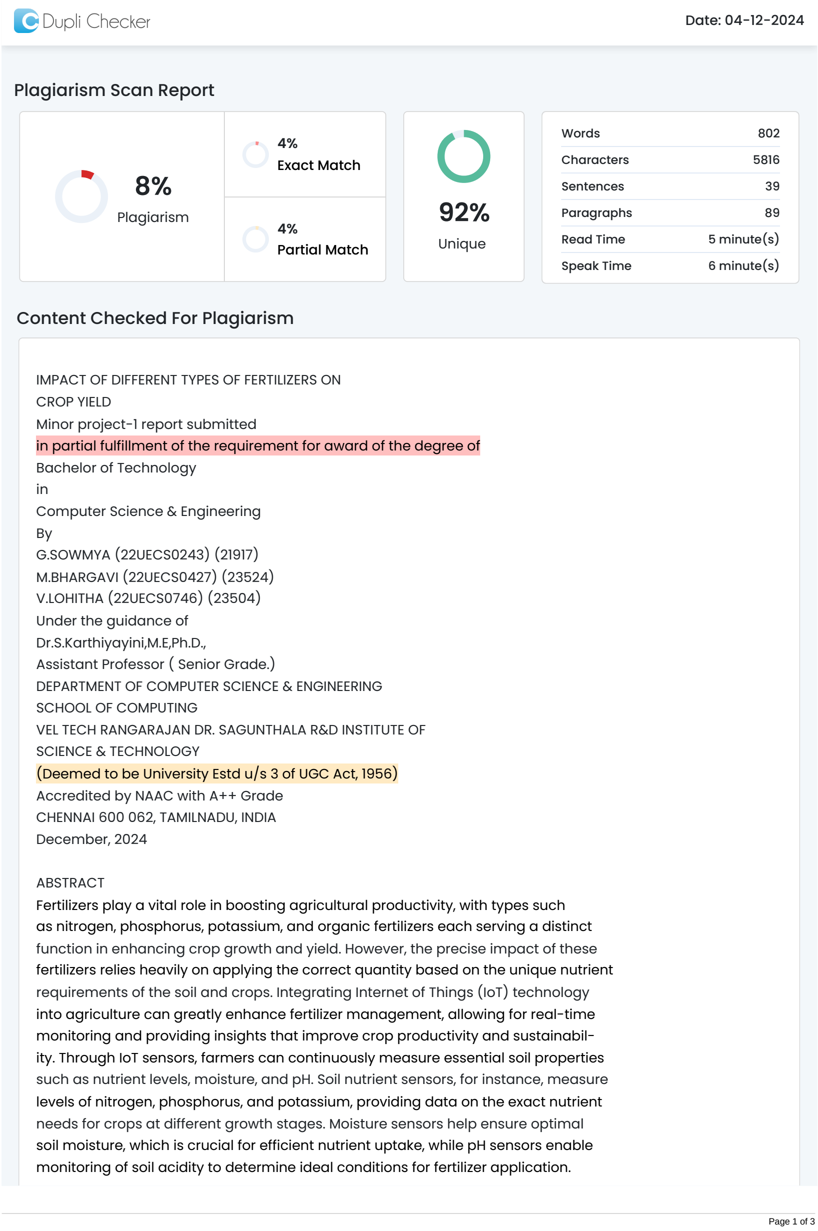
7.2 Future Enhancements

Future enhancements for the impact of different types of fertilizers on crop yield using IoT technology and sensors could focus on expanding the system's capabilities to include more sophisticated data analytics and machine learning algorithms. By integrating advanced predictive models, the system can improve its accuracy in recommending the optimal type and amount of fertilizer based on real-time soil conditions and historical data. Deep learning techniques, for instance, could analyze complex patterns in soil pH, moisture, and nutrient levels to make more precise predictions. Additionally, incorporating a wider range of sensors, such as multi-spectral imaging and remote sensing technologies, could provide more comprehensive data on soil health and crop conditions, enabling even finer adjustments to fertilizer application.

Moreover, developing a user-friendly interface, such as a mobile app or web platform, could greatly enhance the usability of the system. This platform could provide real-time notifications, detailed reports, and actionable insights to farmers, helping them make informed decisions about fertilizer use. Features like predictive analytics for future crop cycles, integration with local weather forecasts, and automated adjustment recommendations based on changing soil conditions could further optimize fertilizer application. Additionally, the system could include blockchain technology to ensure data security and transparency, making the data collected tamper-proof and traceable. Expanding the system's database to include a variety of crops and regional soil conditions could make the technology more versatile, allowing it to be used by farmers in different geographical locations and farming conditions. This would not only improve crop yields and reduce waste but also promote sustainable agricultural practices globally.

Chapter 8

PLAGIARISM REPORT



Dupli Checker

Date: 04-12-2024

Plagiarism Scan Report

8%	Plagiarism
4%	Exact Match
4%	Partial Match
92%	Unique
Words	802
Characters	5816
Sentences	39
Paragraphs	89
Read Time	5 minute(s)
Speak Time	6 minute(s)

Content Checked For Plagiarism

IMPACT OF DIFFERENT TYPES OF FERTILIZERS ON
CROP YIELD
Minor project-1 report submitted
in partial fulfillment of the requirement for award of the degree of
Bachelor of Technology
in
Computer Science & Engineering
By
G.SOWMYA (22UECS0243) (21917)
M.BHARGAVI (22UECS0427) (23524)
V.LOHITHA (22UECS0746) (23504)
Under the guidance of
Dr.S.Karthiyayini,M.E,Ph.D.,
Assistant Professor (Senior Grade.)
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF COMPUTING
VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE & TECHNOLOGY
(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA
December, 2024

ABSTRACT
Fertilizers play a vital role in boosting agricultural productivity, with types such as nitrogen, phosphorus, potassium, and organic fertilizers each serving a distinct function in enhancing crop growth and yield. However, the precise impact of these fertilizers relies heavily on applying the correct quantity based on the unique nutrient requirements of the soil and crops. Integrating Internet of Things (IoT) technology into agriculture can greatly enhance fertilizer management, allowing for real-time monitoring and providing insights that improve crop productivity and sustainability. Through IoT sensors, farmers can continuously measure essential soil properties such as nutrient levels, moisture, and pH. Soil nutrient sensors, for instance, measure levels of nitrogen, phosphorus, and potassium, providing data on the exact nutrient needs for crops at different growth stages. Moisture sensors help ensure optimal soil moisture, which is crucial for efficient nutrient uptake, while pH sensors enable monitoring of soil acidity to determine ideal conditions for fertilizer application.

Page 1 of 3

Figure 8.1: **Plagariarism Report**

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